

Supplemental Material for “Onset of a quantum Fisher–Selke sequence in the transverse-field ANNNI model and its bilayer”

S1. EXACT-DIAGONALIZATION BENCHMARK

Table S1 compares the SSE code with exact diagonalization at finite temperature on four 12-spin periodic clusters built by the production lattice code (Sec. IV C of the main text). The benchmark anneals every case over 16 fields from $g = 3$ to the target, with 2000 sweeps per field, followed by 5000 thermalization and 40 000 measurement sweeps at the target, in 24 independent chains.

S2. PRE-REGISTERED TEST OF THE $\langle 23 \rangle$ PHASE

The test of Sec. V B of the main text followed a written plan. Its design, analysis and pass criteria were fixed before the production runs, after an independent review of the design.

Design. Seven points (κ, r) with $\lambda \leq 0.5$, chosen so that the predicted margin of $\langle 23 \rangle$ at g_c exceeds the expected statistical error. At each point the pairs $\{\langle 23 \rangle, \langle 2 \rangle\}$ ran on a 20×20 bilayer and $\{\langle 23 \rangle, \langle 3 \rangle\}$ on a 20×30 bilayer, with 12 chains per structure, 2000 + 3000 sweeps per field at $T = 0.15 J_0$, and the ascending template protocol of Sec. IV B of the main text. The field grid had steps of 0.1 below the predicted window, steps of 0.02 (or 0.01) across it with a margin of at least 0.06, and g_c as an exact grid point. Competitors ($\langle 223 \rangle$, $\langle 233 \rangle$, $\langle 2333 \rangle$) ran at four of the points, and one point was repeated at $T = 0.10 J_0$.

Analysis. Free energies as in Sec. IV B of the main text. Uncertainties from the jackknife over chains; differences tested with Welch’s t and its Satterthwaite degrees of freedom; crossings from a cubic-spline root of the free-energy difference, with linear interpolation in g^2 as a cross check.

Criteria. (T1) Existence at each point: $F_{\langle 23 \rangle} < F_{\langle 2 \rangle}$ and $F_{\langle 23 \rangle} < F_{\langle 3 \rangle}$ at g_c , each with one-sided $p < 0.0228$, and no chain cut before g_c . The headline required a pass at all seven points, and a failure at $\lambda \leq 0.4$ was to be reported and investigated. (T2) Window edges compared with the exact- κ $O(g^4)$ and $O(g^6)$ values (not pass/fail). (T3, T4) Width of the window and the ratio of widths for $r = 0$ and 1. (T5) Competitors at g_c against their $O(g^6)$ values, with a stop rule: no claim for $\langle 23 \rangle$ if a competitor lay below it at g_c by more than 3σ . (T6) Consistency of $\langle 23 \rangle$ on the two lattices ($\max |z| < 3$) and of $\langle 3 \rangle$ with the 24×24 branches. (T7) Window edges at $T = 0.10$ and 0.15 agree within 2σ . (T8) An exploratory search for further structures above the window.

Table S2 lists the outcomes. The stop rule of T5 was never triggered. At $\kappa = 0.53$, $r = 1$, T6 passed its cri-

terion, but with $\chi^2 = 3.45$ per point and 14 of 25 fields beyond 2σ ; the original 20×30 run for $\langle 23 \rangle$ at this point is the one that the repeat with four times the sweeps did not reproduce (Sec. V B of the main text).

Changes. Before the runs, the review moved $\langle 3 \rangle$ from a 12×12 to the 20×30 lattice, added g_c to the grids, specified the quadrature, and replaced a test on $\min(F_{\langle 2 \rangle}, F_{\langle 3 \rangle})$ by the two separate comparisons of T1. After the first output existed, eight ambiguities in the analysis were resolved in writing without changing the text of the plan, and an error that affected only chains cut at a change of structure was corrected. It had no effect on any chain used in T1. The follow-up at $\kappa = 0.52$, $r = 1$ was decided after the result of T1 at that point was known, and is reported separately.

S3. ADDITIONAL RUNS AND ERROR BUDGET

A. Common lattice, replicates and sweeps

After the pre-registered test (Sec. V B of the main text) we ran the following checks; Table S3 collects the results. For decoupled layers the common lattice at $\kappa = 0.53$ places the upper edge 1.7σ above the eighth-order value and 2.9σ above the sixth. In the bilayer the pre-registered upper edges lie above the series, and the paired differences of $e_b - f_b$ between $\langle 23 \rangle$ and $\langle 3 \rangle$ are consistent with zero, so the integration does not cause this excess.

To locate the bilayer excess we followed all three structures on one 20×60 lattice at both bilayer points, twice with independent seeds, and repeated $\langle 23 \rangle$ and $\langle 3 \rangle$ with four times the sweeps on both lattices at $\kappa = 0.52$ (Table S3). For decoupled layers the common lattice agrees with the pre-registered values and with the eighth-order series at both κ ; at $\kappa = 0.52$ it also turns the narrowest pre-registered margin, $\langle 23 \rangle$ below $\langle 3 \rangle$, into a clear one. In the bilayer the two 20×60 runs agree with each other, and together they place both edges and both gains within 2σ of the eighth-order series. The gain over $\langle 2 \rangle$ is the same on every lattice. The gain over $\langle 3 \rangle$ is about 10^{-4} per spin larger on 20×30 at the standard sweeps, in every set run with these settings: the original runs, the follow-up at $\kappa = 0.52$ and the repeat at $T = 0.10 J_0$. With four times the sweeps it falls to the 20×60 value at both points, while on 20×60 it does not change. At $\kappa = 0.52$ the four combinations of lattice and sweeps thus place the shift in the one combination of 20×30 and standard sweeps (interaction 1.9σ , simple effects 2.1 – 2.5σ), and we think these runs were not fully equilibrated. The original 20×30 run for $\langle 23 \rangle$ at $\kappa = 0.53$ had also been flagged by the consistency check T6 (Appendix C of the

TABLE S1. SSE against exact diagonalization on 12-spin periodic clusters. e is the energy per spin and m_x the transverse magnetization, both in units of J_0 ; the SSE values are means of 24 chains with standard errors in parentheses. The last column is the largest $|z|$ over all observables of the case (31–39 per case).

Cluster	T/J_0	J_y/J_0	κ	$r_\perp$	g	e (ED)	e (SSE)	m_x (ED)	m_x (SSE)	$\max z $
2×3 bilayer	0.15	1	0	1	1.0	-2.60042	-2.60075(20)	0.20169	0.20176(14)	1.65
2×3 bilayer	0.15	1	0.75	-1	0.5	-1.78588	-1.78603(17)	0.14416	0.14394(23)	2.01
2×3 bilayer	0.50	1	0.5	1	2.0	-2.52091	-2.52076(68)	0.54569	0.54629(57)	2.62
2×6 layer	0.15	1	0	0	2.0	-2.52595	-2.52599(23)	0.56144	0.56135(18)	0.68
2×6 layer	0.15	1	0.75	0	1.0	-1.83990	-1.84065(41)	0.46098	0.46162(69)	1.83
2×6 layer	0.15	1	0.5625	0	1.5	-2.07157	-2.07171(41)	0.64328	0.64283(47)	1.86
2×6 layer	0.15	-1	0.25	0	1.0	-1.89600	-1.89616(20)	0.29902	0.29881(20)	1.82
2×6 layer	0.50	1	1.0	0	0.5	-1.71530	-1.71590(69)	0.18847	0.18876(157)	2.00
1×6 ladder	0.15	1	0.75	1	1.0	-1.46156	-1.46128(36)	0.61156	0.61140(57)	1.80
1×6 ladder	0.15	1	1.3	-1	0.3	-1.30261	-1.30361(86)	0.22459	0.22781(527)	2.50
1×6 ladder	0.15	1	0.5	1	3.0	-3.13579	-3.13617(44)	0.95525	0.95528(12)	1.68
3×4 layer	0.15	1	0.375	0	2.5	-2.74079	-2.74034(26)	0.88540	0.88520(14)	1.68
3×4 layer	0.15	1	1.0	0	1.0	-2.12664	-2.12642(20)	0.25671	0.25629(20)	2.08

TABLE S2. Outcomes of the pre-registered criteria. T2: pulls of the lower and upper edges against $O(g^6)$. T5: largest |pull| of a competitor against $O(g^6)$. T6: $\max|z|$ of $\langle 23 \rangle$ on 20×20 against 20×30 , with χ^2 per point. T7: pulls of the edges at $T = 0.10 J_0$ against $T = 0.15 J_0$. T8: fields at which $\langle 233 \rangle$ lies 2σ below $\langle 23 \rangle$ and $\langle 3 \rangle$. T3 and T4 are in Table VI of the main text and Fig. 5(c) of the main text.

κ	r	T1	T2	T5	T6	T7	T8
0.52	0	pass	-1.8, -1.1	—	1.15 (0.57)	—	—
0.53	0	pass	+0.1, +2.7	1.4	1.35 (0.59)	-0.1, -1.7	none
0.545	0	pass	-0.7, +5.1	1.2	1.85 (0.55)	—	none
0.5625	0	pass	-0.9, +11.4	3.3	1.04 (0.34)	—	1.10–1.30
0.52	1	fail	+1.4, +1.2	—	1.38 (1.14)	—	—
0.53	1	pass	-0.5, +2.9	1.5	2.19 (3.45)	—	none
0.545	1	pass	+0.4, +1.4	—	1.15 (0.34)	—	—

main text). The energies shift with the free energies, so the integration is not the cause, and the pair correlations along y show no wall defects. Runs repeated with identical settings and new seeds agree within their statistical errors (rms $t = 1.06$ over ten comparisons). We therefore take the 20×60 and four-times-sweeps values as our estimates. They agree with the eighth-order series within 1σ in the gain over $\langle 3 \rangle$ and the upper edge at both points; in the bilayer the sixth and eighth orders differ by less than these errors. The pre-registered ΔF_3 in the bilayer carries a bias of about -10^{-4} per spin, which raises the upper edge by about 0.008 in g . $\langle 23 \rangle$ lies below both neighbors in every run and on every lattice at all seven points. At $\kappa = 0.52$, $r = 1$ its margin over $\langle 3 \rangle$ is 1.6 – 1.9×10^{-4} per spin in the three unbiased sets, five to nine standard errors, and its margin over $\langle 2 \rangle$ on the common lattice is 1.6×10^{-4} , 6.5 standard errors.

At $T = 0.10 J_0$, where the thermal contributions estimated below are about ten times smaller, $\langle 23 \rangle$ remains below both neighbors at $\kappa = 0.53$ for both r (Table S3). For $r = 0$ the lower edge and the gain over $\langle 2 \rangle$ are unchanged, and the upper edge and the gain over $\langle 3 \rangle$ move by 1.7σ and 1.5σ ; for $r = 1$ both edges and both gains agree with those at $T = 0.15 J_0$ within 1σ .

B. Error budget

The error budget of the free-energy differences at the test points has five parts. The statistical errors are 0.3 – 1.2×10^{-4} per spin (Table VI of the main text). The choice of quadrature (trapezoidal, Richardson or spline) moves them by up to 0.4×10^{-4} . The truncation of the series, estimated by the last retained term, the difference between the eighth- and sixth-order values of Tables VI and VII of the main text, is at most 0.5×10^{-4} in the bilayer and 0.7×10^{-4} for $r = 0$ at $\kappa \leq 0.53$, but 2.6×10^{-4} and 7.6×10^{-4} for ΔF_3 at $\kappa = 0.545$ and 0.5625 , $r = 0$. Thermal excitations of the rows next to the walls contribute at $T = 0.15 J_0$. For $r = 0$ a free-chain estimate, which neglects the pinning of these rows by the walls and therefore bounds the effect from above, gives up to 0.3×10^{-4} at $\kappa = 0.52$, 1.0×10^{-4} at $\kappa = 0.53$ and several 10^{-4} at $\kappa \geq 0.545$; it favors $\langle 23 \rangle$ over $\langle 2 \rangle$ and $\langle 3 \rangle$ over $\langle 23 \rangle$, the opposite of the upper-edge excess. For $r = 1$ the rows form ladders whose kink energy at the test fields exceeds $1.2 J_0$, and the estimate is below 10^{-5} . Finally, the pre-registered 20×30 runs in the bilayer carry the bias of about -10^{-4} in ΔF_3 discussed in Sec. S3 A. For $r = 0$ at $\kappa \geq 0.545$ the comparison of ΔF_3 and of the upper edge with the series is therefore qualitative.

TABLE S3. Test points on different lattices and in different runs: window edges and the free-energy differences ΔF_2 and ΔF_3 of $\langle 23 \rangle$ relative to $\langle 2 \rangle$ and $\langle 3 \rangle$ at g_c (in $10^{-4} J_0$ per spin, negative where $\langle 23 \rangle$ is lower), with statistical errors. The pre-registered runs use 20×20 for $\langle 2 \rangle$ and 20×30 for $\langle 3 \rangle$; the follow-up at $\kappa = 0.52$ was decided afterwards; “4 $\times$ ” has four times the sweeps; 20×60 holds all three structures, “replicate” repeats that run with new seeds, and “pooled” combines the chains of both; “ $T = 0.10$ ” repeats the pre-registered runs at $T = 0.10 J_0$; 40×40 holds $\langle 2 \rangle$ and $\langle 23 \rangle$ only.

κ, r	run	$g_{2 23}$	$g_{23 3}$	ΔF_2	ΔF_3
0.52, 0	pre-registered	0.5885(21)	0.6084(36)	-3.32(60)	-1.46(62)
	20×60	0.5917(13)	0.6130(28)	-2.34(38)	-2.26(47)
	$O(g^8)$	0.5922	0.6134	-2.21	-2.32
0.53, 0	pre-registered	0.7050(19)	0.7553(51)	-4.93(76)	-7.19(99)
	20×60	0.7038(11)	0.7522(37)	-5.41(42)	-6.84(72)
	$T = 0.10$	0.7048(17)	0.7451(33)	-5.12(65)	-5.36(66)
	$O(g^8)$	0.7045	0.7460	-5.16	-5.63
0.53, 1	pre-registered	1.1037(29)	1.1604(27)	-4.31(66)	-4.90(34)
	$20 \times 30, 4\times$	—	1.1550(29)	—	-4.23(37)
	20×60	1.1033(19)	1.1483(39)	-4.38(45)	-3.38(51)
	replicate	1.1017(18)	1.1552(32)	-4.75(40)	-4.25(43)
	pooled	1.1025(13)	1.1517(26)	-4.56(30)	-3.82(34)
	40×40	1.1049(11)	—	-4.03(27)	—
	$T = 0.10$	1.1069(17)	1.1612(37)	-3.56(39)	-5.06(49)
	$O(g^8)$	1.1046	1.1536	-4.08	-4.09
0.52, 1	pre-registered	0.9297(30)	0.9577(52)	-0.99(54)	-2.47(58)
	follow-up	0.9232(25)	0.9620(36)	-2.13(46)	-2.90(39)
	20×60	0.9258(19)	0.9477(40)	-1.67(34)	-1.38(44)
	replicate	0.9269(20)	0.9527(38)	-1.48(35)	-1.90(41)
	pooled	0.9264(13)	0.9502(28)	-1.58(24)	-1.64(30)
	$20 \times 30, 4\times$	—	0.9526(23)	—	-1.91(25)
	$20 \times 60, 4\times$	—	0.9524(20)	—	-1.87(22)
	$O(g^8)$	0.9253	0.9515	-1.78	-1.78